

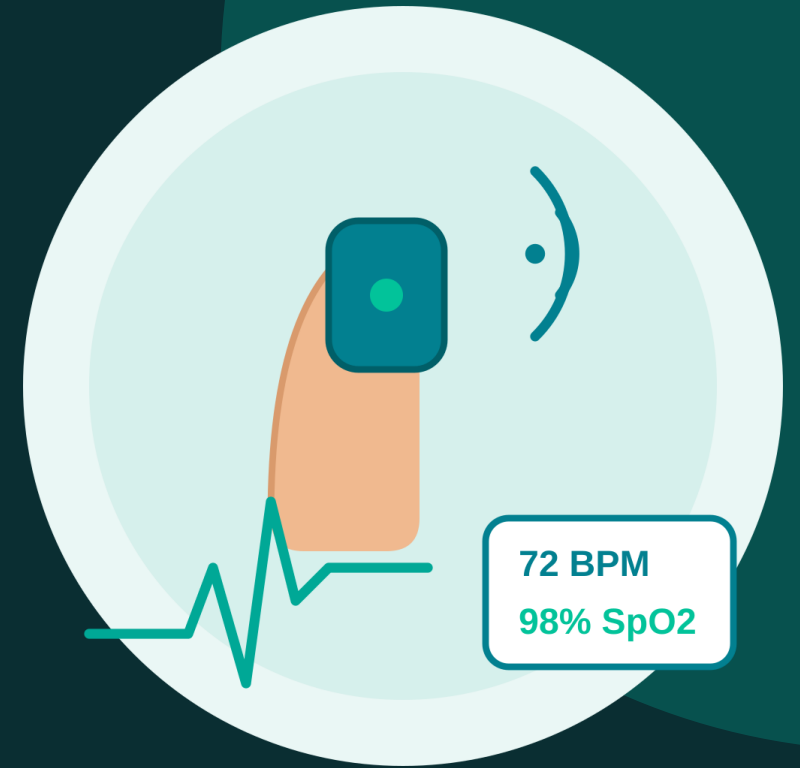
TINKERING PROJECT

VASC Age

Smart Health Monitoring System

School: Annasaheb Dhamane Madhyamik Ashram School, Kaman

Team Size: 5 Students



The Problem

Many elderly people and patients need regular health monitoring, but checking vital signs frequently can be difficult without visiting a hospital or clinic.

- ✓ Frequent hospital visits are time-consuming and costly
- ✓ Elderly and chronically ill patients need continuous tracking
- ✓ Manual checkups can't be done round the clock at home



Continuous

health monitoring shouldn't need
a hospital visit every time

Project Objectives

To design an affordable, portable device that continuously tracks vital signs — helping people monitor their health quickly and accurately, without frequent hospital visits.



Accurate Monitoring

Deliver reliable heart rate and SpO₂ readings in real time.



Portability

Build a compact device usable at home, school, or camps.



Quick Checkups

Cut down time spent on routine hospital visits.



Affordability

Use low-cost components for wide accessibility.

VASC Age – Smart Health Monitoring System



Real-time Heart Rate

Continuously measures pulse (BPM) using an optical sensor.



Blood Oxygen (SpO₂)

Tracks blood oxygen saturation levels instantly.



Instant Data Display

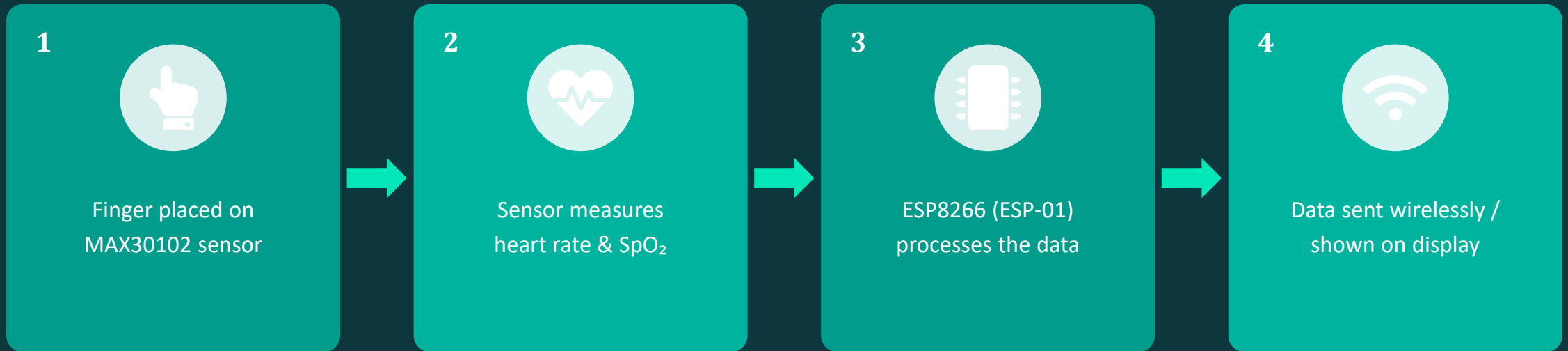
Shows readings clearly on a connected display.



Portable & Wireless

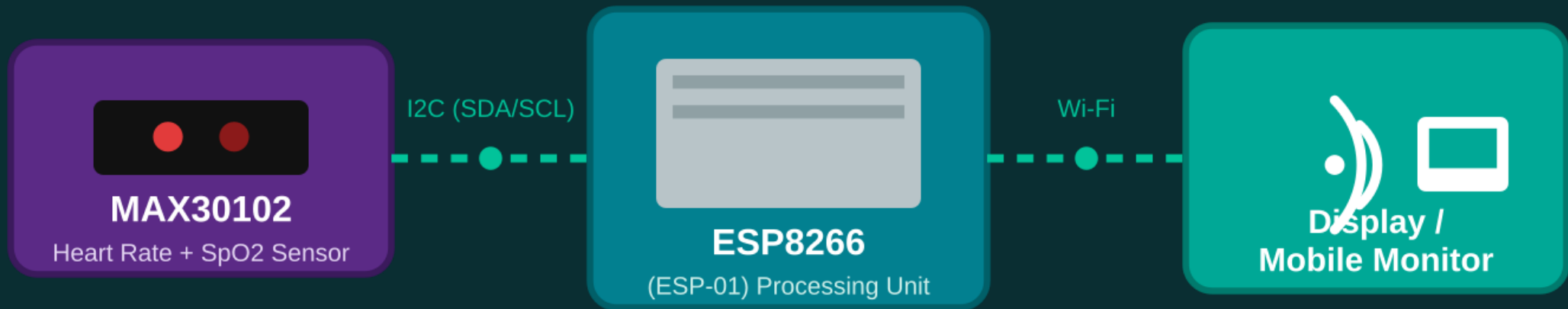
Compact, easy-to-use design for monitoring anywhere.

How It Works



The final health readings help users track their vitals quickly, accurately, and without visiting a clinic.

Circuit Connection Overview



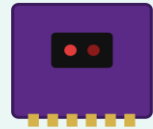
Components Used



ESP-01 ESP8266 Wi-Fi Module

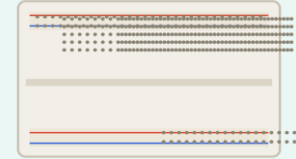
Processes sensor data & enables wireless connectivity

MAX30102 Sensor



MAX30102 Heart Rate & SpO₂ Sensor

Measures pulse rate and blood oxygen level



Breadboard

Breadboard

Solderless board to build the circuit prototype



Jumper Wires

Jumper Wires

Connects components on the breadboard



USB Cable

USB Cable

Powers the circuit and enables programming

Advantages



Provides quick and accurate health readings



Helps in early health monitoring



Suitable for home and healthcare use



Portable and easy to use



Cost-effective solution

Real-Life Uses



Hospitals



Clinics



Homes



Health Camps



Elderly Care Centers



Schools for Health Check-ups



Thank You

VASC Age — Smart Health Monitoring System

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